Statistics and Trends: Exploring London's Weather

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Introduction:

In this report, an analysis of daily weather in London during 1979 to 2020 is being performed. The dataset consists of 15,341 observations involving various weather conditions, like temperature, sunshine, solar radiation, rainfall, and air pressure. The primary purpose of this report is to determine any relationships that exist between these various weather conditions. Some of the key aspects of analysis include daily average temperature, relationship between hours of sunshine and maximum temperature and relationship among various weather measures.

About Dataset:

The London Weather dataset involves daily weather information in London. It consists of various weather conditions and weather measures. The dataset is quite informative in understanding changes in temperature, rainfall, humidity and other weather phenomena.

Data Source and Citation:

The dataset used in this analysis was sourced from Kaggle, which is an online data analysis site that offers a variety of datasets for data analysis and research purposes. I used the London Weather dataset due to the valuable weather information present within. The London Weather dataset can be accessed on the official Kaggle website. Reference : Kaggle , London daily weather, 1979–2020.

1.Distribution of daily mean temperature:

From the histogram above, we can clearly see the distribution of the daily mean temperature in London. The average temperature was 11.48°C whereas the median was 11.4°C, indicating that the temperatures were mainly in the 11-12°C range. A standard deviation of 5.73°C showed that temperatures differed during the period. Majority of the temperatures were in the range of 7.0°C – 16.0°C although the overall temperatures ranged from -7.6°C to 29.0°C.

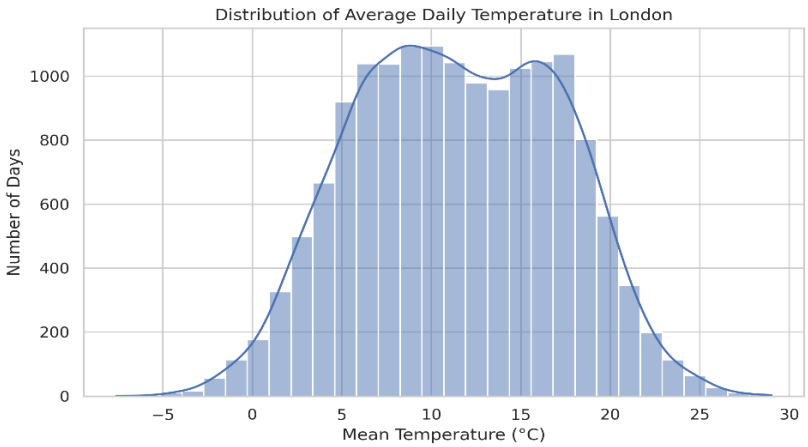


Figure 1. Distribution of daily mean temperature (°C).

2.Sunshine and maximum temperature:

The scatter diagram represents the relationship between sunshine hours per day and maximum temperature. The Pearson correlation coefficient is 0.47, which indicates that there exists a moderately positive relationship between the two factors. This implies that those days with higher sunshine levels are likely to experience higher temperatures. However, the scatter plot also illustrates that sunshine is not the only factor that determines the temperature since the dots are scattered all over the diagram. There are other meteorological factors that may affect the results.

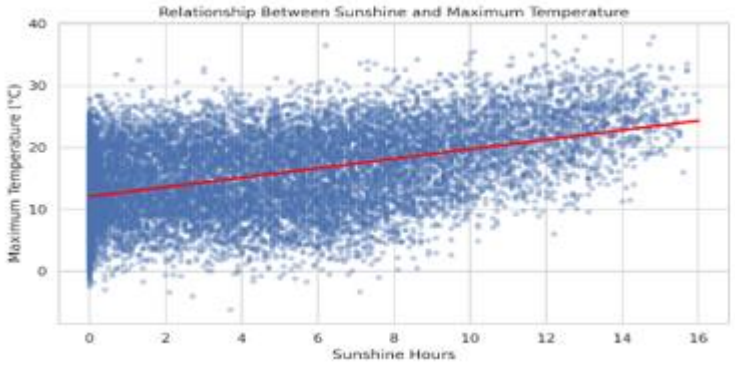


Figure 2. Sunshine hours compared with daily maximum temperature.

3.Correlations among weather factors:

Heat map indicates the correlations among seven numerical weather factors. The maximum temperature and mean temperature are highly positively correlated with each other (r=0.91), which was obvious since both indicate the temperature at the same time. Sunshine factor and Global Radiation factor have a high positive correlation (r=0.85). It can be said that sunny days usually get more energy from the sun. The precipitation factor and air pressure have a negative correlation with each other (r=-0.35), that is, higher air pressure usually means less precipitation.

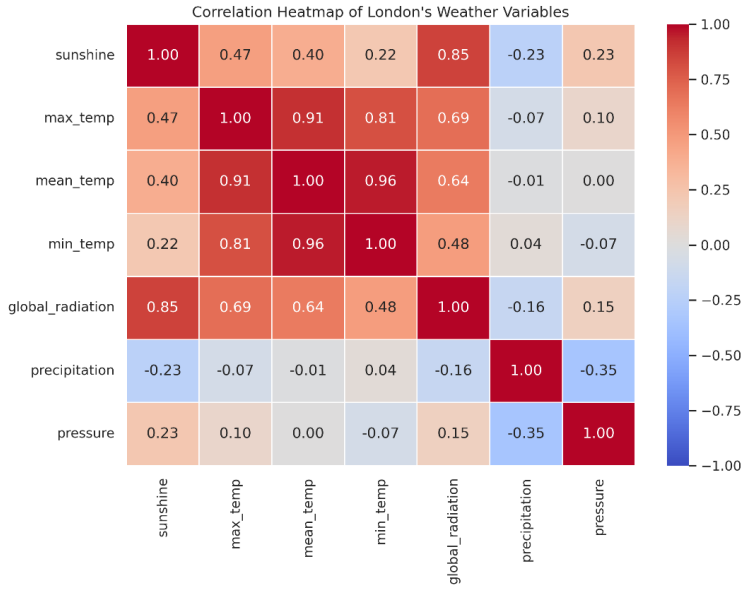


Figure 3. Correlation matrix for selected London weather variables

Conclusion:

Overall, the analysis shows that there is a definite seasonal trend in London's weather, which includes a daily average temperature of 11.48°C. Sunshine was found to have a moderate positive relationship with maximum temperature, and the correlation map clearly showed that the various temperature variables and sunshine had a very high relationship with each other and with solar radiation. This is an example of how several weather variables can be analyzed simultaneously to gain better insights into the climatic trends of London.

GIT-HUB link: https://github.com/Muralimannam/Statistics-Trends-Report-London-Weather-analysis